Tree Chatbot BETA

+ New chat

WORKSPACE

- Product research 6
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RECENT

API rate limits
Weekly planning
Untitled chat

Your keys. Your conversations.
Saved locally in this browser.

JD Personal workspace
Local · no account needed

PROVIDER Gemini | MODEL gemini-2.5-pro

02

key saved ••••1234 Change Remove

Product research > Chat chain

Search architecture

All changes saved locally

Search architecture
Gemini · Root chat

How should we design search for our docs? Compare a few approaches.

ASSISTANT

Start with hybrid search.
Combine exact keyword matches with semantic retrieval.

1. **Vector search** for intent
2. **Reranking** for relevance

Ask a follow-up...

Branch 2 children | Delete

Delete child chats before deleting this chat.

Embeddings

Vector search deep dive
Claude · Branched from Search architecture

Let's explore embeddings. How do we measure similarity?

ASSISTANT

Use cosine similarity.
Embed each document and query into the same vector space.

$$\cos(\theta) = (a \cdot b) / (\|a\| \|b\|)$$

Ask a follow-up...

Branch | Delete

Ranking

Ranking & evaluation
OpenAI · Branched from Search architecture

How do we combine keyword and vector results?

ASSISTANT

Reciprocal rank fusion (RRF)
Merge both ranked lists without normalizing their raw scores.

Start with k = 60, then evaluate on real queries using NDCG@10.

Ask a follow-up...

Branch | Delete

100%

Scroll to zoom · Space + drag to pan

MINIMAP